Explicit Formula for $P_n(x)$

- Using this recurrence, the Legendre polynomial of degree n is

$$P_n(x) = \sum_{k=0}^{m} (-1)^k \frac{(2n-2k)!}{2^n\, k!\,(n-k)!\,(n-2k)!}\, x^{n-2k}, \qquad m = \lfloor n/2 \rfloor.$$

$$= \frac{(2n)!}{2^n (n!)^2} x^n - \frac{(2n-2)!}{2^n\, 1!\,(n-1)!(n-2)!} x^{n-2} + \cdots$$

- Though $P_n(x)$ is a polynomial, it is treated as a **special function**, because of its many remarkable properties.



Rewriting the Sum via Differentiation

- Notice that for $0 \leq k \leq m = \lfloor n/2 \rfloor$,

$$\frac{d^n}{dx^n} x^{2n-2k} = \frac{(2n-2k)!}{(n-2k)!} x^{n-2k}.$$

- So $P_n(x)$ can be rewritten as

$$P_n(x) = \sum_{k=0}^{m} (-1)^k \frac{(2n-2k)!}{2^n\, k!\, (n-k)!\, (n-2k)!} x^{n-2k}$$

$$= \frac{1}{2^n n!} \frac{d^n}{dx^n} \sum_{k=0}^{m} (-1)^k \frac{n!}{k!(n-k)!} x^{2n-2k}.$$



Toward Rodrigues' Formula

- When $k > m = \lfloor n/2 \rfloor$, each term in the sum above is a polynomial of degree *less than* n, so its nth derivative is 0.
- So the sum can be extended from $m+1$ up to n, without changing its value:

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}\sum_{k=0}^{n}(-1)^k \frac{n!}{k!(n-k)!}x^{2n-2k} = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n.$$

- The last equality follows from the Binomial expansion of $(x^2-1)^n$.



Rodrigues' Formula (Method 1: From the Explicit Sum)

- We thus obtain one of the most important identities for Legendre polynomials:

$$\boxed{P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n}$$

- This is called **Rodrigues' Formula**.
- It gives $P_n(x)$ directly, without needing the series recursion — differentiate $(x^2 - 1)^n$ exactly n times, then divide by $2^n n!$.
- It is also the starting point for proving many other properties of $P_n(x)$, including orthogonality.



Proof that $P_n(1) = 1$

- Recall Leibniz's rule: $\dfrac{d^n(fg)}{dx^n} = \displaystyle\sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \frac{d^k f}{dx^k} \frac{d^{n-k} g}{dx^{n-k}}$.
- In Rodrigues' formula, write $(x^2-1)^n = (x+1)^n (x-1)^n$ and apply Leibniz's rule:

$$P_n(x) = \frac{1}{2^n n!} \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \frac{d^k[(x+1)^n]}{dx^k} \frac{d^{n-k}[(x-1)^n]}{dx^{n-k}}.$$

- The $k = 0$ term is $(x+1)^n \cdot n!$; every other term still contains a factor of $(x-1)$ (since fewer than n derivatives are taken of $(x-1)^n$).
- Evaluating at $x = 1$, every term except $k = 0$ vanishes. Hence

$$P_n(1) = \frac{1}{2^n n!}(1+1)^n n! = 1.$$



Rodrigues' Formula: A Second, Independent Derivation

- The derivation just given builds Rodrigues' formula *up* from the explicit series for $P_n(x)$.
- There is a completely independent route: start from the candidate expression $z = (x^2 - 1)^n$, show that $u = d^n z/dx^n$ satisfies Legendre's equation directly, and separately fix the normalizing constant using $P_n(1) = 1$.
- This "Method 2" is self-contained — it does not need the explicit series formula at all.
- This formulation helps to prove orthogonality, compute norms, and derive many other identities for Legendre polynomials.



Method 2: Setting Up the ODE for $z = (x^2 - 1)^n$

- **State and prove Rodrigues' Formula.**
- Let $z = (x^2 - 1)^n$. Then $z' = 2nx(x^2 - 1)^{n-1}$, so

$$(x^2 - 1)z' = 2nx\, z. \tag{2}$$

- Differentiate (2) exactly $(n + 1)$ times using Leibniz's rule (HOMEWORK). Most terms combine, leaving

$$(1 - x^2)\frac{d^{n+2}z}{dx^{n+2}} - 2x\frac{d^{n+1}z}{dx^{n+1}} + n(n + 1)\frac{d^n z}{dx^n} = 0. \tag{3}$$

- Writing $u = \dfrac{d^n z}{dx^n}$, Equation (3) becomes exactly

$$(1 - x^2)u'' - 2xu' + n(n + 1)u = 0, \tag{4}$$

which is **Legendre's equation**! So $u = d^n z/dx^n$ solves it, meaning $u = C \cdot P_n(x)$ for some constant C as P_n is solution of LDE.



Method 2: Fixing the Normalizing Constant

- So $\dfrac{d^n}{dx^n}(x^2-1)^n = C\,P_n(x)$ for some constant C; we now find C using $P_n(1) = 1$.
- Write $z = (x^2-1)^n = (x+1)^n(x-1)^n$ and differentiate n times by Leibniz's rule:

$$\frac{d^n z}{dx^n} = (x+1)^n \cdot n! + (\text{terms that still contain a factor of } (x-1)).$$

- Every term except the first vanishes at $x = 1$; the first term at $x = 1$ gives $(x+1)^n \cdot n!\big|_{x=1} = 2^n\, n!$.
- So $\left.\dfrac{d^n z}{dx^n}\right|_{x=1} = 2^n\, n!$. Since $P_n(1) = 1$, this forces $C = 2^n\, n!$.
- Therefore, the **Rodrigues' Formula** is

$$\boxed{P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n.}$$



Problem : Computing $P_4(x)$ and $P_5(x)$

- Use Rodrigues' formula to compute $P_4(x)$ and $P_5(x)$.
- By definition, $P_4(x) = \dfrac{1}{2^4\,4!}\dfrac{d^4}{dx^4}(x^2-1)^4$.
- Expanding $(x^2-1)^4 = x^8 - 4x^6 + 6x^4 - 4x^2 + 1$ and differentiating 4 times, then dividing by $2^4 \cdot 24 = 384$:

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3).$$

- Similarly, $P_5(x) = \dfrac{1}{2^5\,5!}\dfrac{d^5}{dx^5}(x^2-1)^5$ gives

$$P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x).$$

- Check: $P_4(1) = \frac{1}{8}(35 - 30 + 3) = 1$ and $P_5(1) = \frac{1}{8}(63 - 70 + 15) = 1$, as required.



Generating Function: Setting Up

- **State and prove Generating function for Legendre polynomials**
- By the Binomial theorem, $(1+z)^r = \sum_{n=0}^{\infty} \frac{r(r-1)\cdots(r-n+1)}{n!} z^n$ for $|z| < 1$.
- Take $z = t^2 - 2xt = t(t-2x)$, and assume $|t^2 - 2xt| < 1$. With $r = -\frac{1}{2}$:

$$(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} \frac{(-\frac{1}{2})(-\frac{3}{2})\cdots(-\frac{1}{2}-n+1)}{n!} t^n (t-2x)^n$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{2^{2n}(n!)^2} t^n (t-2x)^n.$$

- Expanding $(t-2x)^n$ by the Binomial theorem and simplifying:

$$(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} \sum_{k=0}^{n} (-1)^k \frac{(2n)!}{2^{2n} n!\, k!\, (n-k)!} t^{n+k} (2x)^{n-k}.$$



Generating Function: Simplifying the Double Sum

- Using the reindexing identity

$$\sum_{n=0}^{\infty}\sum_{k=0}^{n} C_{k,n}t^{n+k} = \sum_{n=0}^{\infty}\sum_{k=0}^{\lfloor n/2\rfloor} C_{k,n-k}\, t^{n}:$$

$$(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty}\sum_{k=0}^{\lfloor n/2\rfloor} (-1)^k \frac{(2n-2k)!}{2^n\, k!\,(n-k)!\,(n-2k)!}\, t^n x^{n-2k}.$$

- The inner sum is exactly our earlier closed-form expression for $P_n(x)$.



Generating Function: Final Result

- We conclude that

$$\boxed{(1 - 2xt + t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x)\, t^n}$$

- That is, $(1 - 2xt + t^2)^{-1/2}$ is the **generating function** of $\{P_n(x)\}$.
- This single closed-form expression encodes *every* Legendre polynomial as a Taylor coefficient in t.



Analyticity of the Generating Function

- Let $G(x,t) = (1 - 2xt + t^2)^{-1/2}$.
- Note that $1 - 2xt + t^2 = (t - x)^2 + (1 - x^2) > 0$ for every real t, whenever $|x| < 1$.
- So $G(x,t)$ is analytic (as a function of t) for every real t, whenever $x \in (-1, 1)$.
- Hence $G(x,t)$ can be expanded as a power series in t around $t = 0$:

$$G(x,t) = \sum_{n=0}^{\infty} P_n(x)\, t^n. \qquad (*)$$

- This justifies term-by-term manipulation of the series — a key technical point for the results that follow.



Consequence: $P_n(1) = 1$ and $P_n(-1) = (-1)^n$

- Take $x = 1$ in $(*)$:

$$\frac{1}{\sqrt{1-2t+t^2}} = \frac{1}{1-t} = \sum_{n=0}^{\infty} P_n(1)\, t^n \quad \Rightarrow \quad P_n(1) = 1.$$

- Take $x = -1$ in $(*)$:

$$\frac{1}{\sqrt{1+2t+t^2}} = \frac{1}{1+t} = \sum_{n=0}^{\infty} P_n(-1)\, t^n \quad \Rightarrow \quad P_n(-1) = (-1)^n.$$

- Both give a second, quick confirmation of results proved earlier by other means.



Consequence: Values at $x = 0$

- Take $x = 0$ in $(*)$: $(1 + t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(0)\, t^n$.
- Comparing this with the Binomial expansion of $(1 + t^2)^{-1/2}$ (which contains only even powers of t) gives

$$P_{2n+1}(0) = 0, \qquad P_{2n}(0) = (-1)^n \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n\, n!}.$$

- So every Legendre polynomial of odd degree vanishes at the origin — consistent with odd-degree P_n being an odd function.



P_0–P_5 via the Generating Function

- **Q.** Using the generating function directly, verify $P_0, \ldots, P_5$.
- Expand $(1 - 2xt + t^2)^{-1/2} = \left[1 - t(2x - t)\right]^{-1/2}$ by the Binomial series in powers of t, and simultaneously expand $\sum_n P_n(x)t^n$.
- Equating coefficients of $t^0, t^1, t^2, t^3, t^4, t^5$ on both sides directly reproduces

$$P_0 = 1,\ P_1 = x,\ P_2 = \tfrac{1}{2}(3x^2 - 1),\ P_3 = \tfrac{1}{2}(5x^3 - 3x),$$

$$P_4 = \tfrac{1}{8}(35x^4 - 30x^2 + 3),\ P_5 = \tfrac{1}{8}(63x^5 - 70x^3 + 15x).$$

- This gives an alternative, purely algebraic way to generate the polynomials, without solving the ODE directly.



Example: A Formula for $P_n(0)$

- **Q.** Prove: $P_n(0) = 0$ if n is odd, and $P_{2m}(0) = \dfrac{(-1)^m(2m)!}{2^{2m}(m!)^2}$.
- Put $x = 0$ in the generating function: $\sum_n P_n(0)t^n = (1+t^2)^{-1/2}$.
- The right side, expanded by the Binomial theorem, contains **only even** powers of t — so every odd-indexed coefficient must vanish: $P_{2m+1}(0) = 0$.
- Equating the coefficient of t^{2m} on both sides gives

$$P_{2m}(0) = \frac{1\cdot 3\cdot 5\cdots(2m-1)}{2\cdot 4\cdots(2m)}(-1)^m = \frac{(-1)^m(2m)!}{2^{2m}(m!)^2}.$$

- E.g. $m = 2$: $P_4(0) = \dfrac{4!}{2^4(2!)^2} = \dfrac{24}{64} = \dfrac{3}{8}$ — matching the explicit formula for $P_4(x)$.



Example: The Generating Function Solves Legendre's Equation

- **Q.** Prove that $z = (1 - 2xt + t^2)^{-1/2}$, written as $z = \sum_n t^n P_n(x)$, satisfies $(1 - x^2)z_{xx} - 2xz_x$ term-by-term matching the Legendre operator applied to each P_n.
- Term by term, $(1 - x^2)z_{xx} - 2xz_x = \sum_n t^n \left[(1 - x^2)P_n'' - 2xP_n'\right]$.
- Since each P_n satisfies Legendre's equation, $(1 - x^2)P_n'' - 2xP_n' = -n(n+1)P_n$, the left side collapses to $-\sum_n n(n+1)P_n(x)t^n$ — the same combination that appears when Legendre's equation is applied to each term of the series, confirming z satisfies the associated identity term by term.
- This is a useful consistency check linking the generating function directly back to the defining ODE.

